Oral Pathology Study Guide (Chapter 8)

Focused Review for NBDHE Preparation (Questions 1 to 50 & Clinical Cases A & B)

How to use this guide: Study the yellow "High-Yield" boxes first. They contain the core pathology facts tested in Questions 1 to 50 and Clinical Cases A & B. Then use the question-to-concept map and clinical vocabulary table at the end to master your terminology.

---

1. Soft Tissue Lesions & Viral Infections (HPV & Varicella-Zoster Virus)

- **Fordyce Granules**: Ectopic sebaceous glands appearing as white-to-yellow raised spots on hairless mucosal surfaces. Present in most adults on the buccal mucosa and vermilion border. **Appear during puberty due to stimulation by androgenic hormones** (not present in pre-pubertal children).
- **Oral Lichen Planus**: Chronic mucocutaneous disorder. The most common clinical form is **reticular lichen planus**, characterized by white lacy lines (**Wickham striae**). The erosive form presents with painful, ulcerated gingival lesions.
- **Varicella-Zoster Virus (VZV)**: Reactivation of latent VZV causes **Shingles**. Herpes Simplex Virus Type 1 (HSV-1) causes herpes labialis and primary herpetic gingivostomatitis. Coxsackievirus causes herpangina.
- **Leukoedema vs. Morsicatio Buccarum (Cheek Biting)**: Leukoedema is caused by intracellular edema and **disappears when the tissue is stretched** (bluish opalescent appearance). Cheek biting exhibits macerated tissue and does NOT disappear upon stretching.
- **Irritation Fibroma (Traumatic Fibroma)**: Benign reactive exophytic hyperplasia of fibrous connective tissue caused by **chronic trauma**. Most common site: **buccal mucosa along the occlusal plane**, lips, tongue, and gingiva. Most common in **45-year-old females**.
- **Oral Squamous Papilloma & Verruca Vulgaris**: Benign exophytic epithelial growths caused by **Human Papillomavirus (HPV)** featuring a **"cauliflower-like" appearance** (finger-like projections). HPV types 6 & 11 cause papillomas; HPV-2 & 4 cause verruca vulgaris.
- **Oropharyngeal HPV & Cancer**: HPV (especially HPV-16) is a major oncogenic risk factor for tonsillar/oropharyngeal carcinoma. Higher prevalence in **males**. Palpation of cervical lymph nodes and salivary ducts is critical because HPV has an affinity for salivary gland ductal epithelium.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 1, 2, 3, 4, 5, 6, 7, 25, 38, 39, 40, 48, 50 & CASE A:** * **Fordyce granules** are white-yellow sebaceous glands in the buccal mucosa; respond to **androgenic hormones at puberty**. * **Reticular lichen planus** exhibits a white lacy pattern (Wickham striae) and is the most common type. * Reactivation of varicella-zoster virus causes **Shingles**. * **Leukoedema** disappears when the buccal tissue is stretched. * **Irritation fibroma** is caused by chronic bite trauma (most common in **45-year-old females**). * Oropharyngeal HPV lesions have a **cauliflower-like appearance**, are more common in **males**, and HPV clears in an average of **12 months**. * **Hairy tongue** is caused by elongation of **filiform papillae**.

---

2. Salivary Gland Pathology & Associated Lesions

- **Mucocele (Mucus Extravasation Phenomenon)**: **Pseudocyst** (not a true cyst because it is **NOT lined by epithelium**; the mucin pool is surrounded by **granulation tissue**). Caused by traumatic severance of a **minor** accessory salivary gland duct (e.g., lip biting).

- **Most common site**: **Lower lip** (lateral to midline).
- **Clinical presentation**: Soft, fluctuant, moveable, non-painful, fluid-filled nodule.
- **Ranula**: Clinical term for a large mucocele located in the **floor of the mouth**, involving the sublingual or submandibular gland ducts (Rivinus or Wharton ducts).
- **Linea Alba**: Horizontal white keratinized line along the buccal mucosa at the occlusal plane, caused by occlusal friction.
- **Nicotine Stomatitis**: Hard palate lesion caused by heat and tobacco smoke. Presents as white keratinized mucosa dotted with **prominent red puncta** (inflamed minor salivary gland duct openings).
- **Necrotizing Sialometaplasia**: Benign lesion featuring ischemic necrosis of palatal minor salivary glands, secondary to **vascular ischemia from trauma or local anesthetic injection**.
- **Pleomorphic Adenoma (Benign Mixed Tumor)**: Most common benign salivary gland tumor (**90% of benign salivary tumors**). Occurs most frequently in the **parotid gland** in adults (ages 35–55). Has a 5% malignant transformation rate.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 8, 9, 10, 11, 17, 20 & CASE B:** * A **mucocele** is NOT a true cyst because mucin is **surrounded by granulation tissue**, not epithelium. * Most common site for a mucocele is the **lower lip**; presents as a non-painful, moveable fluid-filled nodule. * A mucocele in the floor of the mouth is termed a **Ranula**. * **Nicotine stomatitis** occurs on the **hard palate** (inflamed minor salivary ducts). * **Necrotizing sialometaplasia** is caused by **ischemia or local anesthetic trauma**. * **Pleomorphic adenoma** is the most common salivary tumor and occurs in the **parotid gland**.

---

3. Odontogenic & Non-Odontogenic Cysts of Head and Neck

- **Radicular (Periapical) Cyst**: The **most common odontogenic cyst overall** (60–70%). Forms at the apex of a non-vital tooth following pulp necrosis and inflammation.
- **Dentigerous (Follicular) Cyst**: The **second most common odontogenic cyst**. A developmental cyst forming around the crown of an unerupted or impacted tooth (frequently **mandibular third molars**).
- **Residual Cyst**: Persists after incomplete removal of a tooth that had a radicular cyst.
- **Simple Bone Cyst (Traumatic Bone Cyst)**: Intraosseous pseudocyst lacking an epithelial lining. Radiographically displays a radiolucency with a **scalloped margin encircling tooth roots**. Treatment: **surgical curettage** to induce bleeding and bone repair within 6 months.
- **Thyroglossal Tract Cyst**: Developmental cyst located in the **midline of the neck**. Most common in young adults (<20 years). Primary clinical evaluation: **thorough palpation examination of the thyroid gland**.
- **Nasolabial Cyst**: Developmental soft tissue cyst located in the mucolabial fold near the maxillary canine and base of the nose.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 18, 19, 21, 22, 23, 46, 47, 49:** * **Periapical/radicular cyst** is the most common odontogenic cyst. * **Dentigerous cyst** surrounds the crown of an unerupted tooth (third molars) and is **developmental** (not inflammatory). * **Residual cyst** results from incomplete removal of a previous tooth/cyst. * **Simple bone cyst** displays a **scalloped appearance around roots** and is treated with **curettage**. * **Thyroglossal tract cyst** is in the **midline of the neck** and requires a thyroid exam.

---

4. Developmental Tooth Anomalies & Bone Structures

- **Anodontia & Ectodermal Dysplasia**: Ectodermal dysplasia is an inherited disorder featuring **congenital absence of teeth (anodontia/hypodontia)**, frequently premolars and incisors, along with hair, nail, and sweat gland defects.
- **Concrescence**: Developmental anomaly where two adjacent teeth are **united solely by root cementum**.
- **Fusion**: Union of two developing tooth germs by dentin, resulting in a large tooth with a single or divided root canal.
- **Gemination**: Single tooth germ attempting to divide, resulting in a bifid crown with one root and one canal.
- **Dens in Dente (Dens Invaginatus)**: Enamel organ invagination into the dental papilla. Most common location: **maxillary lateral incisors**.
- **Natal Teeth**: Teeth present at birth (frequently primary mandibular central incisors with undeveloped, unstable roots).
- **Amelogenesis Imperfecta**: Inherited enamel defect divided into 4 types. The **hypocalcified type** is a defect in enamel matrix mineralization.
- **Abfraction**: Wedge-shaped cervical tooth defect at the **Cementoenamel Junction (CEJ)** caused by flexural occlusal forces.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 14, 16, 34, 35, 36, 37, 42:** * **Ectodermal dysplasia** causes congenital absence of premolars and other teeth. * **Concrescence** is the union of two roots by **cementum** only. * **Dens invaginatus (dens in dente)** is most frequent in **maxillary lateral incisors**. * **Natal teeth** are present at birth with undeveloped, unstable roots. * **Abfraction** occurs at the **cervical area of teeth at the CEJ**.

---

5. Systemic Bone Pathology & Hereditary/Blood Disorders

- **Torus Palatinus & Mandibularis**: Benign bony exostoses. Torus palatinus occurs in the midline of the hard palate. Torus mandibularis is most common in **young males aged 13 to 16 years** on the lingual premolar surface.
- **Paget Disease of Bone**: Bone metabolic disorder characterized by disorganized osteoclast/osteoblast activity. Most common in **Caucasian males over 50 years** (NOT African American males). Causes maxillary enlargement, tooth spacing, and ill-fitting dentures.
- **Osteomalacia & Osteopenia**: Osteomalacia ("soft bones") in adults is due to vitamin D/calcium deficiency. **Osteopenia** is the term for progressive bone demineralization and reduced density in adults over time.
- **Hyperparathyroidism**: Endocrine disorder featuring hypercalcemia, **Brown tumors**, kidney stones, bone resorption, and loss of lamina dura.
- **Cherubism**: Autosomal dominant disorder (SH3BP2 gene mutation) featuring bilateral symmetric jaw expansion in **very young children (ages 1 to 5)**.
- **Von Willebrand Disease & Hemophilia**: Von Willebrand disease is an autosomal dominant clotting factor deficiency affecting **males and females**. Hemophilia is X-linked.
- **Iron Deficiency Anemia**: Clinical signs include mucosal pallor, **angular cheilitis**, **loss of filiform papillae (atrophic glossitis)**, and tongue burning.
- **Plummer-Vinson Syndrome**: Associated with iron deficiency anemia, glossitis, and dysphagia.
- **Chronic Myelogenous Leukemia (CML)**: Characterized by the **Philadelphia 22 chromosome** genetic translocation.

- **Addison Disease**: Primary adrenal insufficiency producing melanocyte stimulation and **diffuse brown macular oral pigmentation**.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 12, 13, 24, 26, 27, 29, 30, 31, 33, 41, 43, 44, 45:** * **Iron deficiency anemia** causes loss of filiform papillae, angular cheilitis, and tongue burning. * **Philadelphia 22 chromosome** is associated with **chronic leukemia**. * **Cherubism** occurs in **very young children** (ages 1–5) due to SH3BP2 mutation. * **Osteopenia** refers to adult bone demineralization over time. * **Hyperparathyroidism** causes Brown tumors, kidney stones, and loss of lamina dura. * **Von Willebrand disease** affects clotting in both males and females. * **Addison disease** causes oral melanotic pigmentation. * **Paget disease** is most common in **Caucasian males**.

---

6. Clinical Case Study Reviews (Cases A & B)

Case A: Patient Jeff (Oropharyngeal Pathology & Dysphagia)

- **History**: Jeff is a 30-year-old male hotel manager who lived abroad for several years without dental care. Lost teeth from **boxing without a mouthguard**. Has caries and 5 mm periodontal pockets.
- **Chief Complaint**: Complains of **throat fullness when swallowing** and **dysphagia** (difficulty swallowing) for 6 months **without pain**. Initially noticed redness; now sees a thick white plaque.
- **Oral Exam**: Ulcerative, white plaque-like **"cauliflower-like" mass** in posterior oropharynx blocking left tonsillar tissue.
- **Diagnosis & Management**:
- Primary concern is **HPV-associated Squamous Papilloma / Squamous Cell Carcinoma**.
- HPV is most prevalent in **males** and has a **75% to 85%** early cure rate.
- Immediate action: **Urgent referral to an Otolaryngologist (ENT physician)** for **biopsy**.

Case B: Lower Lip Nodule (Mucocele)

- **History & Exam**: Blue, raised, moveable, non-painful nodule on the lower lip (Fig. 8.91).
- **Histology**: Spilled mucin pool walled off by **granulation tissue without epithelial lining** (Fig. 8.92).
- **Diagnosis**: **Mucocele** caused by traumatic rupture of a minor accessory salivary gland duct.

---

7. Question-to-Concept Map (Questions 1 to 50)

Question	Topic Evaluated	Key Fact to Recall
Q1	Fordyce Granules	White-yellow sebaceous glands in buccal mucosa.
Q2	Reticular Lichen Planus	Most common type; presents with white lacy Wickham striae.
Q3	Varicella-Zoster Virus	Reactivation causes **Shingles**.

Q4	Leukoedema	Disappears when buccal mucosa is stretched.
Q5	Androgenic Hormones	Fordyce granules develop at puberty due to androgenic stimulation.
Q6	Irritation Fibroma	Fibrous hyperplasia caused by chronic bite trauma.
Q7	Fibroma Demographics	Most common in adults aged 40–60 (**45-year-old female**).
Q8	Mucocele Definition	Not a true cyst because mucin is **not lined by epithelium**.
Q9	Ranula	Large mucocele in the floor of the mouth affecting sublingual/submandibular glands.
Q10	Linea Alba	White keratinized line on buccal mucosa at occlusal plane.
Q11	Nicotine Stomatitis	Hard palate lesion with inflamed red minor salivary duct openings.
Q12	Torus Palatinus Etiology	Caused by genetics and mastication; **bacteria/viruses are NOT etiologic**.
Q13	Torus Mandibularis	Most common in **young males aged 13 to 16 years**.
Q14	Ectodermal Dysplasia	Inherited disorder causing congenital absence of premolars/teeth.
Q15	Hairy Tongue	Xylitol consumption does NOT cause hairy tongue; radiation, alcohol, and antibiotics do.
Q16	Concrescence	Two teeth joined solely by root **cementum**.
Q17	Pleomorphic Adenoma	Accounts for **90%** of benign salivary gland tumors.
Q18	Thyroglossal Tract Cyst	Primary clinical exam is **palpation of the thyroid gland**.
Q19	Thyroglossal Cyst Age	Commonly diagnosed in **young adults (<20 years)**.

Q20	Pleomorphic Adenoma Site	Occurs most frequently in the **parotid gland** (NOT submandibular).
Q21	Most Common Odontogenic Cyst	**Radicular (periapical) cyst** is the most common odontogenic cyst.
Q22	Residual Cyst	Caused by **incomplete removal** of a tooth/radicular cyst.
Q23	Simple Bone Cyst	Treated with **surgical curettage** to induce bleeding and bone repair.
Q24	Sickle Cell Anemia	Prevalent in African American and **Mediterranean/Asian** populations.
Q25	Oropharyngeal Cancer	**Human Papillomavirus (HPV)** is the major viral risk factor.
Q26	Plummer-Vinson Syndrome	Associated with iron deficiency anemia and dysphagia.
Q27	Iron Deficiency	Causes loss of filiform papillae, angular cheilitis, and burning tongue.
Q28	Oral HIV Implication	**Oral candidiasis** is the most common oral manifestation (85%).
Q29	Philadelphia 22 Chromosome	Genetic marker for **Chronic Myelogenous Leukemia (CML)**.
Q30	Cherubism Age	Diagnosed in **very young children** (ages 1 to 5).
Q31	Cherubism Risk	Risk of developing **sarcoma** / aggressive giant cell lesions.
Q32	Mucous Pemphigoid	Requires evaluation to prevent severe **eye damage** (conjunctival scarring).
Q33	Osteopenia	Term for adult bone demineralization and reduced density over time.
Q34	Abfraction	Wedge-shaped cervical defect at the **CEJ** caused by occlusal flexure.
Q35	Amelogenesis Imperfecta	**Hypocalcified type** is a defect in enamel matrix mineralization.
Q36	Concrescence	Union of two roots during cementum formation.

Q37	Dens Invaginatus	Most common in **maxillary lateral incisors**.
Q38	Geographic Tongue	Benign condition with **NO malignancy component**.
Q39	Fordyce Granules Site	Most common intraoral site is the **buccal mucosa**.
Q40	Hairy Tongue	Caused by elongation of **filiform papillae**.
Q41	Addison Disease	Primary adrenal insufficiency presenting with **oral melanotic pigmentation**.
Q42	Natal Teeth	Teeth present at birth with undeveloped, unstable roots.
Q43	Paget Disease	Most prevalent in **Caucasian males** (NOT African American males).
Q44	Hyperparathyroidism	Causes **Brown tumors**, kidney stones, bone resorption, and loss of lamina dura.
Q45	Von Willebrand Disease	Autosomal dominant bleeding disorder affecting males and females.
Q46	Dentigerous Cyst	**Developmental** cyst surrounding the crown of an unerupted tooth (NOT inflammatory).
Q47	Nasolabial Cyst	Soft tissue developmental cyst in mucolabial fold near canine.
Q48	HPV Clearance	Typical clearance time for HPV is **12 months**.
Q49	Simple Bone Cyst Radiograph	Displays a **scalloped radiolucent margin** around tooth roots.
Q50	Lichen Planus Type	**Reticular type** is the most common clinical form.

8. Clinical Vocabulary (English - Spanish)

Clinical English Term	Spanish Translation	Clinical Definition / Significance

Dysphagia	Disfagia	Difficulty or discomfort in swallowing.
Dysphonia	Disfonía	Hoarseness or abnormal voice quality.
Cauliflower-like appearance	Aspecto de coliflor	Raised, verrucous, irregular tissue growth pattern.
Fordyce granules	Gránulos de Fordyce	Ectopic sebaceous glands in oral mucosa.
Mucocele	Mucocele	Pseudocyst caused by severed minor salivary duct; mucin walled by granulation tissue.
Ranula	Ránula	Large mucocele in floor of mouth (sublingual/submandibular glands).
Wickham striae	Estrías de Wickham	White lacy lines characteristic of reticular lichen planus.
Athletic mouthguard	Protector bucal deportivo	Protective appliance preventing traumatic tooth avulsions in sports.
Biopsy	Biopsia	Removal of tissue sample for diagnostic microscopic examination.
Otolaryngologist (ENT)	Otorrinolaringólogo	Medical specialist for ear, nose, and throat diseases.
Concrescence	Concrescencia	Union of two adjacent teeth by root cementum only.
Dens invaginatus	Dens invaginatus (Dens in dente)	Enamel organ invagination into dental papilla; common in maxillary lateral incisors.
Abfraction	Abfracción	Wedge-shaped cervical defect at the CEJ from flexural forces.
Dentigerous cyst	Quiste dentígero	Developmental cyst surrounding crown of unerupted tooth.
Radicular cyst	Quiste radicular (periapical)	Most common inflammatory odontogenic cyst at apex of non-vital tooth.
Cherubism	Querubismo	Autosomal dominant jaw expansion in very young children (ages 1-5).
Brown tumor	Tumor pardo	Lytic bone lesion characteristic of hyperparathyroidism.

Philadelphia chromosome	Cromosoma Filadelfia	Translocation between chromosomes 9 and 22 diagnostic for CML.
Atrophic glossitis	Glositis atrófica	Loss of filiform papillae associated with iron deficiency anemia.
Necrotizing sialometaplasia	Sialometaplasia necrotizante	Benign ischemic necrosis of palatal minor salivary glands following trauma/anesthesia.

9. Rapid Recall High-Yield Facts

- **Fordyce Granules** → Sebaceous glands in buccal mucosa stimulated by **androgenic hormones at puberty**.
- **Reticular Lichen Planus** → Most common type; features **Wickham striae** (white lacy lines).
- **Leukoedema** → Bluish opalescence that **disappears when buccal mucosa is stretched**.
- **Irritation Fibroma** → Chronic bite trauma lesion; most common in **45-year-old females**.
- **Oropharyngeal HPV** → **Cauliflower-like mass**; more common in **males**; clears in **12 months**.
- **Mucocele** → **Pseudocyst** (no epithelium, lined by **granulation tissue**); most common site: **lower lip**.
- **Ranula** → Large mucocele in the **floor of the mouth**.
- **Nicotine Stomatitis** → **Hard palate** lesion with red inflamed minor salivary duct openings.
- **Pleomorphic Adenoma** → Most common benign salivary tumor (**90%**); located in **parotid gland**.
- **Radicular Cyst** → **Most common odontogenic cyst** (apex of non-vital tooth).
- **Dentigerous Cyst** → Surrounds crown of unerupted tooth; **developmental** cyst.
- **Simple Bone Cyst** → Shows **scalloped radiolucent margin** around roots; treated with **curettage**.
- **Thyroglossal Tract Cyst** → Located in **midline of neck**; requires **thyroid exam**.
- **Ectodermal Dysplasia** → Causes **congenital absence of premolars/teeth**.
- **Concrescence** → Union of two roots by **cementum** only.
- **Dens in Dente** → Most frequent in **maxillary lateral incisors**.
- **Abfraction** → Cervical defect at **CEJ** caused by occlusal flexure.
- **Iron Deficiency Anemia** → Causes **loss of filiform papillae**, angular cheilitis, and burning tongue.
- **Philadelphia 22 Chromosome** → Genetic marker for **Chronic Myelogenous Leukemia (CML)**.
- **Cherubism** → Bilateral jaw expansion in **very young children (ages 1–5)**.
- **Hyperparathyroidism** → Causes **Brown tumors**, kidney stones, and loss of lamina dura.
- **Case A (Jeff)** → Dysphagia + throat fullness + cauliflower mass in oropharynx = **Urgent referral to ENT for biopsy**.